Lesson 4: Method of Moments and Maximum Likelihood Estimation

BSTA 551: Statistical Inference

Jessica Minnier

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Lesson 4: Constructing Estimators

Review: Where We Left Off

Key concepts from Lesson 3:

- MVUE: Best unbiased estimator (smallest variance)
- Likelihood function and log-likelihood
- Fisher Information: $I(\theta) = E[-\ell''(\theta)]$
- Cramér-Rao Lower Bound: $\mathrm{Var}(T) \geq 1/I_n(\theta)$
- Efficient estimators achieve the CRLB

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Today's Goals:

- Learn to construct estimators systematically
- Method of Moments: Simple, intuitive approach
- Maximum Likelihood: Powerful, principled approach
- Compare the two methods

Motivating Question

We've learned how to **evaluate** estimators (bias, variance, efficiency).

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2. **Maximum Likelihood (MLE):** Find parameter that makes data most "likely"

Key Insight

Both methods give us systematic recipes for deriving estimators from any specified distribution.

Method of Moments

Population vs. Sample Moments

Definitions (Devore 7.2)

Population Moments:

- 1st moment: $E(X) = \mu_1'$
- 2nd moment: $E(X^2) = \mu_2'$
- kth moment: $E(X^k) = \mu_k'$

Sample Moments:

- 1st sample moment: $\frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$
- 2nd sample moment: $\frac{1}{n}\sum_{i=1}^{n} X_i^2$
- kth sample moment: $\frac{1}{n}\sum_{i=1}^{n} X_i^k$

The Method of Moments Principle

Method of Moments Estimators (MMEs)

To estimate m unknown parameters $\theta_1, \ldots, \theta_m$:

1. Express the first m population moments as functions of the parameters
2. Set each population moment equal to its sample counterpart
3. Solve the resulting system of equations for $\theta_1, \ldots, \theta_m$

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The idea: If the population and sample moments should be "close," we can estimate parameters by equating them.

Example 1: Exponential Distribution (One Parameter)

Setup: $X_1, \dots, X_n \sim \text{Exponential}(\lambda)$ where λ is the rate parameter.

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Step 1: Population moment: $E(X) = 1/\lambda$

Step 2: Set equal to sample moment: $\frac{1}{\lambda} = \bar{X}$

Step 3: Solve: $\hat{\lambda}_{MME} = \frac{1}{\bar{X}}$

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Intuition

If the population mean is $1/\lambda$, and we estimate the population mean with $\bar{X}$, then we estimate λ with $1/\bar{X}$.

Medical Application: Hospital Wait Times

Emergency room wait times are often modeled with an exponential distribution.

```r
# Wait times (minutes) for 40 patients
wait_times <- c(12, 45, 8, 23, 67, 15, 31, 42, 19, 56,
                28, 11, 38, 22, 49, 17, 33, 25, 41, 14,
                52, 9, 36, 27, 44, 18, 29, 47, 21, 35,
                13, 39, 24, 51, 16, 32, 46, 20, 37, 26)

# Method of moments estimate for λ (per minute)
lambda_hat <- 1 / mean(wait_times)

cat("Sample mean wait time:", round(mean(wait_times), 2), "minutes\n")
```

```
Sample mean wait time: 30.45 minutes
```

```r
cat("MME of λ:", round(lambda_hat, 4), "per minute\n")
```

```
MME of λ: 0.0328 per minute
```

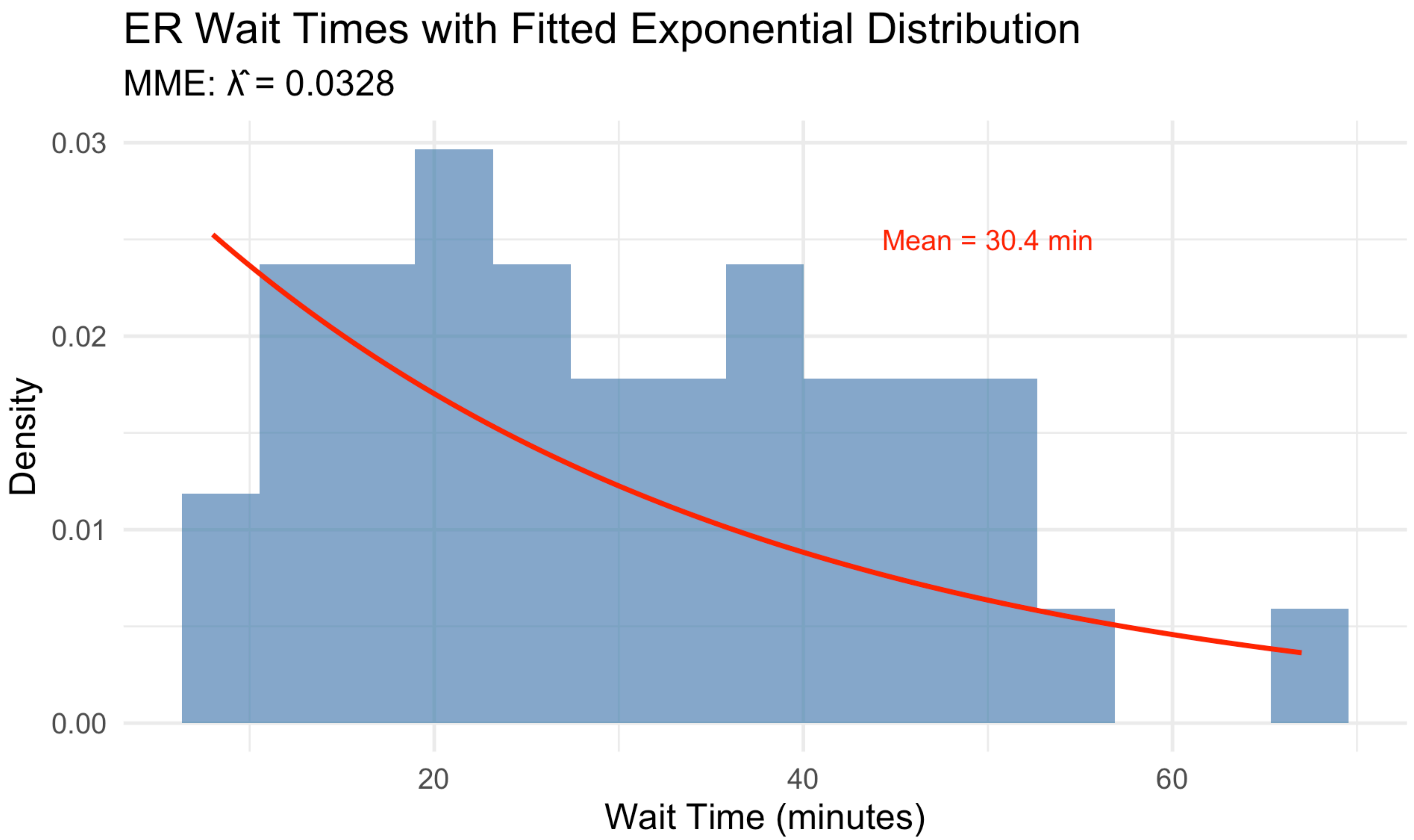
```r
cat("MME of λ:", round(lambda_hat * 60, 2), "per hour")
```

```
MME of λ: 1.97 per hour
```

Visualizing the Exponential Fit: Code

```r
# Compare empirical distribution to fitted exponential
tibble(wait_time = wait_times) |>
  ggplot(aes(x = wait_time)) +
  geom_histogram(aes(y = after_stat(density)), bins = 15,
                 fill = "steelblue", alpha = 0.7) +
  stat_function(fun = dexp, args = list(rate = lambda_hat),
                color = "red", linewidth = 1.2) +
  labs(title = "ER Wait Times with Fitted Exponential Distribution",
       subtitle = str_glue("MME: λ̂ = {round(lambda_hat, 4)}"),
       x = "Wait Time (minutes)", y = "Density") +
  annotate("text", x = 50, y = 0.025,
           label = str_glue("Mean = {round(mean(wait_times), 1)} min"),
           color = "red", size = 5)
```

Visualizing the Exponential Fit: Result



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Setup: $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$

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- $E(X^2) = \mu^2 + \sigma^2$ (variance shortcut formula!)

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Set equal to sample moments:

- $\mu = \bar{X}$
- $\mu^2 + \sigma^2 = \frac{1}{n} \sum X_i^2$

Solving for the Normal MMEs

From the two equations:

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Note on Bias

The MME for σ^2 divides by n, not $n-1$. This means the MME is **biased**!

$$E(\hat{\sigma}^2_{MME}) = \frac{n-1}{n}\sigma^2$$

The usual sample variance S^2 with $n-1$ in the denominator is unbiased.

Example 3: Gamma Distribution (Two Parameters)

Setup: $X_1, \ldots, X_n \sim \text{Gamma}(\alpha, \beta)$

Moments:

- $E(X) = \alpha\beta$
- $E(X^2) = \alpha(\alpha + 1)\beta^2$

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Solutions (after algebra):

$$\hat{\alpha} = \frac{\bar{X}^2}{\frac{1}{n}\sum X_i^2 - \bar{X}^2} = \frac{\bar{X}^2}{\hat{\sigma}^2} \qquad \hat{\beta} = \frac{\frac{1}{n}\sum X_i^2 - \bar{X}^2}{\bar{X}} = \frac{\hat{\sigma}^2}{\bar{X}}$$

Medical Application: Surgery Recovery Times

Recovery times after surgery often follow a gamma distribution.

```r
# Recovery times (days) for 25 patients
recovery_times <- c(4.2, 6.8, 5.1, 8.3, 7.2, 4.9, 6.1, 9.5, 5.7, 7.8,
                    6.4, 8.1, 5.3, 7.5, 6.9, 4.5, 8.7, 5.9, 7.1, 6.2,
                    9.2, 5.5, 7.4, 6.6, 8.4)

# Calculate sample moments
x_bar <- mean(recovery_times)
second_moment <- mean(recovery_times^2)

# MME for gamma parameters
alpha_hat <- x_bar^2 / (second_moment - x_bar^2)
beta_hat <- (second_moment - x_bar^2) / x_bar

cat("Sample mean:", round(x_bar, 2), "days\n")
```

```
Sample mean: 6.77 days
```

```r
cat("MME of α:", round(alpha_hat, 2), "\n")
```

```
MME of α: 22.56
```

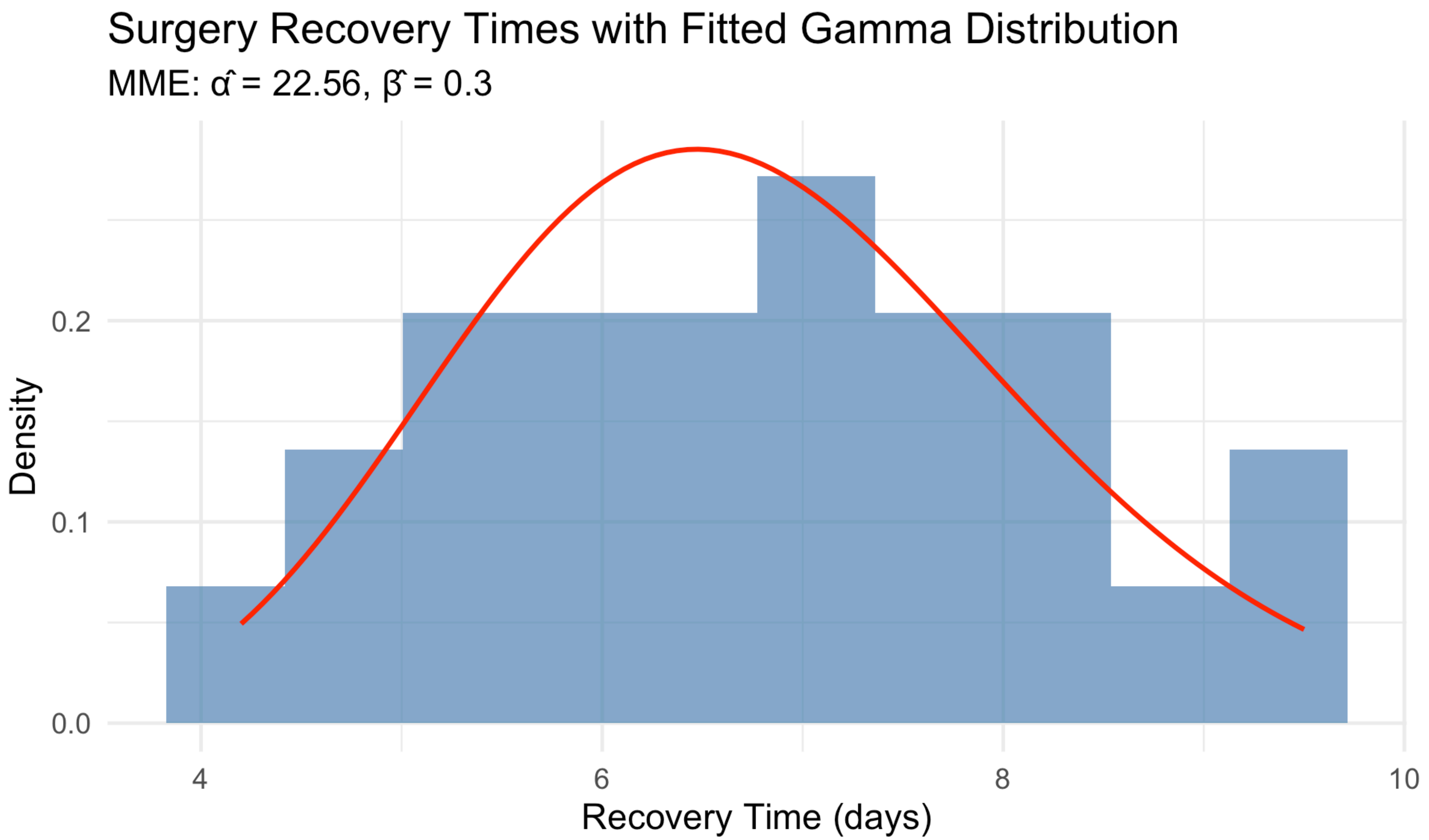
```r
cat("MME of β:", round(beta_hat, 2), "days")
```

```
MME of β: 0.3 days
```

Visualizing the Gamma Fit: Code

```
tibble(recovery_time = recovery_times) |>
  ggplot(aes(x = recovery_time)) +
  geom_histogram(aes(y = after_stat(density)), bins = 10,
                 fill = "steelblue", alpha = 0.7) +
  stat_function(fun = dgamma,
                args = list(shape = alpha_hat, scale = beta_hat),
                color = "red", linewidth = 1.2) +
  labs(title = "Surgery Recovery Times with Fitted Gamma Distribution",
       subtitle = str_glue("MME: α̂ = {round(alpha_hat, 2)}, β̂ = {round(beta_hat, 2)}"),
       x = "Recovery Time (days)", y = "Density")
```

Visualizing the Gamma Fit: Result



Your Turn: Method of Moments

Exercise: Drug concentrations in blood plasma follow a Uniform$[0, \theta]$ distribution. From $n = 15$ measurements, you observe $\bar{x} = 4.2$ mg/L.

Find the MME of θ.

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Solution:

For Uniform[0, θ]: $E(X) = \theta/2$

Setting equal to sample mean: $\bar{x} = \theta/2$

Therefore: $\hat{\theta}_{MME} = 2\bar{x} = 2(4.2) = 8.4$ mg/L

Properties of Method of Moments Estimators

Advantages:

- Simple to compute (usually just algebra)
- Often has closed-form solutions
- Usually consistent (converges to true value as $n \to \infty$)

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- Can sometimes give impossible values
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> **Coming Up Next**
>
> Maximum likelihood estimators are often more efficient than MMEs.

Maximum Likelihood Estimation

Motivating Example: Vaccine Efficacy

A clinical trial for a new vaccine enrolls 200 participants. After exposure to the pathogen, 156 remain disease-free.

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n <- 200
successes <- 156
# What value of p makes this outcome most "likely"?
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Intuition: We should choose the value of p that makes our observed data "most likely" to have occurred.

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This is the essence of **Maximum Likelihood Estimation**.

From Probability to Likelihood

Probability vs. Likelihood: A Key Distinction

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Probability (parameters fixed, data varies):

- Given a known parameter value θ, how likely are different data outcomes?
- Example: If $\theta = 0.8$ for X Binomial(200,θ), what's $P(X = 156)$?

From Probability to Likelihood

Probability vs. Likelihood: A Key Distinction

Probability (parameters fixed, data varies):

- Given a known parameter value θ, how likely are different data outcomes?
- Example: If $\theta = 0.8$ for X Binomial(200,θ), what's $P(X = 156)$?

Likelihood (data fixed, parameters vary):

- Given our observed data, how "likely" are different parameter values?
- Example: We observed 156 successes. Which value of p best explains this?

The Likelihood Function: Definition

Definition (Devore 7.2)

Let $X_1, \ldots, X_n$ have a joint distribution that depends on parameter θ. The **likelihood function** $L(\theta)$ is this joint distribution, regarded as a function of θ (with observed data held fixed).

For a random sample:

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

where f is the pmf (discrete) or pdf (continuous).

Visualizing Likelihood: Code

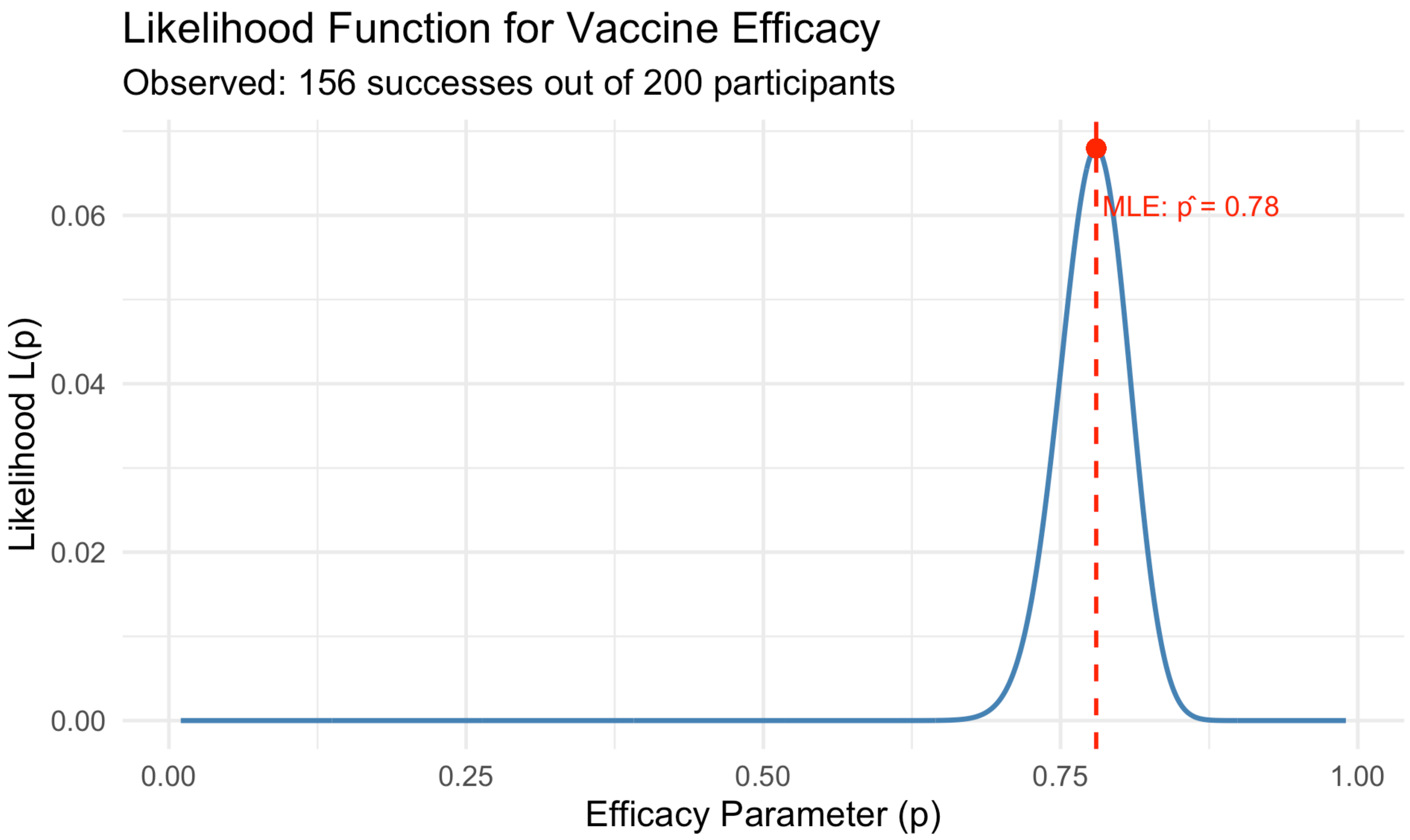
```r
# Likelihood function for vaccine trial
n <- 200
x <- 156

likelihood_data <- tibble(
  p = seq(0.01, 0.99, by = 0.001),
  likelihood = dbinom(x, n, p)
)

mle_p <- x / n

likelihood_data |>
  ggplot(aes(x = p, y = likelihood)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_vline(xintercept = mle_p, color = "red", linetype = "dashed", linewidth = 1) +
  geom_point(aes(x = mle_p, y = dbinom(x, n, mle_p)), color = "red", size = 4) +
  labs(title = "Likelihood Function for Vaccine Efficacy",
       subtitle = str_glue("Observed: {x} successes out of {n} participants"),
       x = "Efficacy Parameter (p)", y = "Likelihood L(p)") +
  annotate("text", x = mle_p + 0.08, y = dbinom(x, n, mle_p) * 0.9,
           label = str_glue("MLE: p̂ = {mle_p}"), color = "red", size = 5)
```

Visualizing Likelihood: Result



The MLE is the value of p at the **peak** of the likelihood function.

Maximum Likelihood Estimate: Definition

Definition

The **Maximum Likelihood Estimate (MLE)** $\hat{\theta}$ is the value of θ that maximizes the likelihood function:

$$\hat{\theta} = \arg\max_{\theta} L(\theta)$$

The MLE is the parameter value that makes the observed data "most likely."

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Notation:

- $\hat{\theta}$ is the MLE (estimator, a random variable)
- $\hat{\theta}$ is also the observed mle (estimate, a specific number from data)

Why Maximum Likelihood?

Intuitive Appeal:

- Choose the parameter that "best explains" our data
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Theoretical Properties (more details later):

1. **Consistent:** $\hat{\theta} \to \theta$ as $n \to \infty$
2. **Asymptotically unbiased:** $E(\hat{\theta}) \approx \theta$ for large n
3. **Asymptotically efficient:** Achieves smallest possible variance for large n
4. **Invariance:** If $\hat{\theta}$ is MLE of θ, then $g(\hat{\theta})$ is MLE of $g(\theta)$

The Log-Likelihood Function

Why Use Log-Likelihood?

The likelihood function is often a **product** of many terms:

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

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Key property: Since $\ln$ is monotonically increasing:

$$\arg\max_{\theta} L(\theta) = \arg\max_{\theta} \ell(\theta)$$

Maximizing the log-likelihood gives the same answer!

Advantages of Log-Likelihood

Computational:

- Sums are easier to differentiate than products
- Avoids numerical underflow (very small products)

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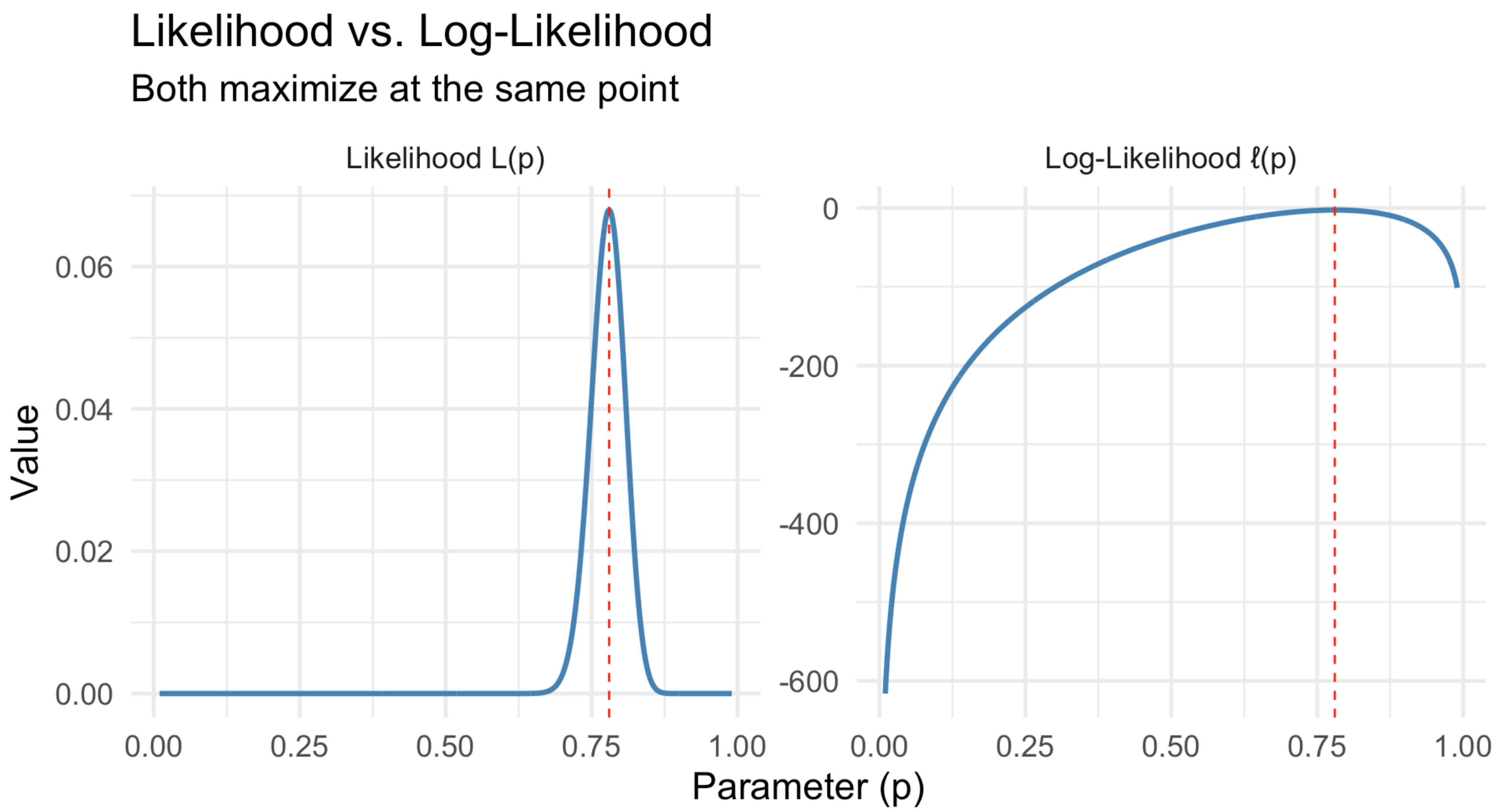
Mathematical:

- $\ell'(\theta) = 0$ is often easier to solve
- Second derivative test is more tractable

Likelihood vs. Log-Likelihood: Code

```r
# Compare likelihood and log-likelihood
likelihood_data |>
  mutate(log_likelihood = log(likelihood)) |>
  select(p, likelihood, log_likelihood) |>
  pivot_longer(-p, names_to = "type", values_to = "value") |>
  mutate(type = case_when(
    type == "likelihood" ~ "Likelihood L(p)",
    type == "log_likelihood" ~ "Log-Likelihood ℓ(p)"
  )) |>
  ggplot(aes(x = p, y = value)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_vline(xintercept = mle_p, color = "red", linetype = "dashed") +
  facet_wrap(~type, scales = "free_y") +
  labs(title = "Likelihood vs. Log-Likelihood",
       subtitle = "Both maximize at the same point",
       x = "Parameter (p)", y = "Value")
```

Likelihood vs. Log-Likelihood: Result



The General MLE Recipe

Finding the MLE

1. **Write down the likelihood:** $L(\theta) = \prod f(x_i; \theta)$
2. **Take the logarithm:** $\ell(\theta) = \sum \ln f(x_i; \theta)$
3. **Differentiate:** Find $\ell'(\theta) = \frac{d\ell}{d\theta}$
4. **Solve:** Set $\ell'(\theta) = 0$ and solve for $\hat{\theta}$
5. **Verify:** Check that $\ell''(\hat{\theta}) < 0$ (maximum, not minimum)

MLE Examples

Example 1: Bernoulli Distribution

Setup: $X_1, \ldots, X_n \sim \mathrm{Bernoulli}(p)$ (iid)

Let $x = \sum x_i$ be the number of successes.

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Step 2 - Log-likelihood:

$$\ell(p) = x\ln(p) + (n-x)\ln(1-p)$$

Example 1: Bernoulli (continued)

Step 3 - Differentiate:

$$\ell'(p) = \frac{x}{p} - \frac{n-x}{1-p}$$

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Step 4 - Solve:

$$\frac{x}{p} = \frac{n-x}{1-p}$$

$$x(1-p) = p(n-x)$$

$$x - xp = np - xp$$

$$x = np$$

$$\hat{p} = \frac{x}{n}$$

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The MLE is the sample proportion! This matches our intuition.

Simulation: Verifying the Bernoulli MLE

```r
true_p <- 0.65
n <- 50
n_sims <- 5000

# Simulate MLE for many samples
bernoulli_sim <- tibble(sim = 1:n_sims) |>
  mutate(
    data = map(sim, \(s) rbinom(n, 1, true_p)),
    mle = map_dbl(data, mean)
  )

# Check properties
bernoulli_sim |>
  summarize(
    mean_mle = mean(mle),
    var_mle = var(mle),
    theoretical_var = true_p * (1 - true_p) / n
  )
```

```
# A tibble: 1 × 3
  mean_mle var_mle theoretical_var
     <dbl>   <dbl>           <dbl>
1    0.650 0.00450         0.00455
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bernoulli_sim <- tibble(sim = 1:n_sims) |>
  mutate(
    data = map(sim, \(s) rbinom(n, 1, true_p)),
    mle = map_dbl(data, mean)
  )

# Check properties
bernoulli_sim |>
  summarize(
    mean_mle = mean(mle),
    var_mle = var(mle),
    theoretical_var = true_p * (1 - true_p) / n
  )
```

```
# A tibble: 1 × 3
  mean_mle var_mle theoretical_var
     <dbl>   <dbl>           <dbl>
1    0.650 0.00450         0.00455
```

Example 2: Exponential Distribution

Setup: $X_1, \ldots, X_n \sim \text{Exponential}(\lambda)$

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Same as MME! This is common for exponential family distributions.

Example 3: Normal Distribution

Setup: $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, both unknown

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Partial derivatives:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum(x_i - \mu) = 0 \Rightarrow \hat{\mu} = \bar{X}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(x_i - \mu)^2 = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{X})^2$$

Important Note on Normal MLE

$$\hat{\sigma}^2_{MLE} = \frac{1}{n} \sum (x_i - \bar{X})^2$$

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⚠ **Bias Alert!**

The MLE divides by n, not $n - 1$:

$$E(\hat{\sigma}^2_{MLE}) = \frac{n-1}{n}\sigma^2 \neq \sigma^2$$

The MLE is **biased** for σ^2!

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The MLE is **biased** for σ^2!

However, for large n, this bias becomes negligible:

$$\frac{n-1}{n} \to 1 \text{ as } n \to \infty$$

Numerical Optimization for MLEs

When Calculus Isn't Enough

Not all MLEs have closed-form solutions!

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Examples requiring numerical optimization:

- Gamma distribution (shape and rate parameters)
- Weibull distribution
- Beta distribution
- Mixed models, GLMs, many real-world applications

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The General Approach

1. Write down the log-likelihood function $\ell(\theta)$
2. Use numerical optimization to find $\hat{\theta} = \arg\max \ell(\theta)$
3. In R: minimize the **negative** log-likelihood (since `optim` minimizes)

Example: Exponential MLE — Calculus vs. Numerical

Let's verify our calculus result numerically:

```r
# Generate exponential data
set.seed(42)
exp_data <- rexp(100, rate = 0.5)  # True λ = 0.5

# Calculus solution: λ̂ = 1/X̄
mle_calculus <- 1 / mean(exp_data)

# Numerical solution: minimize negative log-likelihood
neg_log_lik_exp <- function(lambda, data) {
  if (lambda <= 0) return(Inf)  # λ must be positive
  -sum(dexp(data, rate = lambda, log = TRUE))
}

# Use optimize() for single-parameter optimization
result <- optimize(neg_log_lik_exp, interval = c(0.01, 5), data = exp_data)
mle_numerical <- result$minimum
```

Example: Exponential MLE — Calculus vs. Numerical (solution)

```r
cat("Calculus MLE:  λ̂ =", round(mle_calculus, 5), "\n")
```

```
Calculus MLE:  λ̂ = 0.44472
```

```r
cat("Numerical MLE: λ̂ =", round(mle_numerical, 5), "\n")
```

```
Numerical MLE: λ̂ = 0.44472
```

```r
cat("Difference:", round(abs(mle_calculus - mle_numerical), 8))
```

```
Difference: 2.64e-06
```

They match! ✓

Example: Gamma MLE — Numerical Only

The Gamma distribution has **no closed-form MLE** for the shape parameter α.

Log-likelihood for Gamma(α, β):

$$\ell(\alpha, \beta) = n\alpha \ln(\beta) - n \ln \Gamma(\alpha) + (\alpha - 1) \sum \ln(x_i) - \beta \sum x_i$$

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Setting $\partial\ell/\partial\alpha = 0$ involves the **digamma function** $\psi(\alpha)$ — no closed-form solution!

Gamma MLE: Numerical Optimization in R (1/2)

```r
# Generate gamma data (true α = 3, β = 2)
set.seed(123)
gamma_data <- rgamma(150, shape = 3, rate = 0.5)  # rate = 1/scale

# Define negative log-likelihood for Gamma
neg_log_lik_gamma <- function(params, data) {
  alpha <- params[1]
  beta <- params[2]   # rate parameter
  if (alpha <= 0 | beta <= 0) return(Inf)
  -sum(dgamma(data, shape = alpha, rate = beta, log = TRUE))
}
```

Gamma MLE: Numerical Optimization in R (2/2)

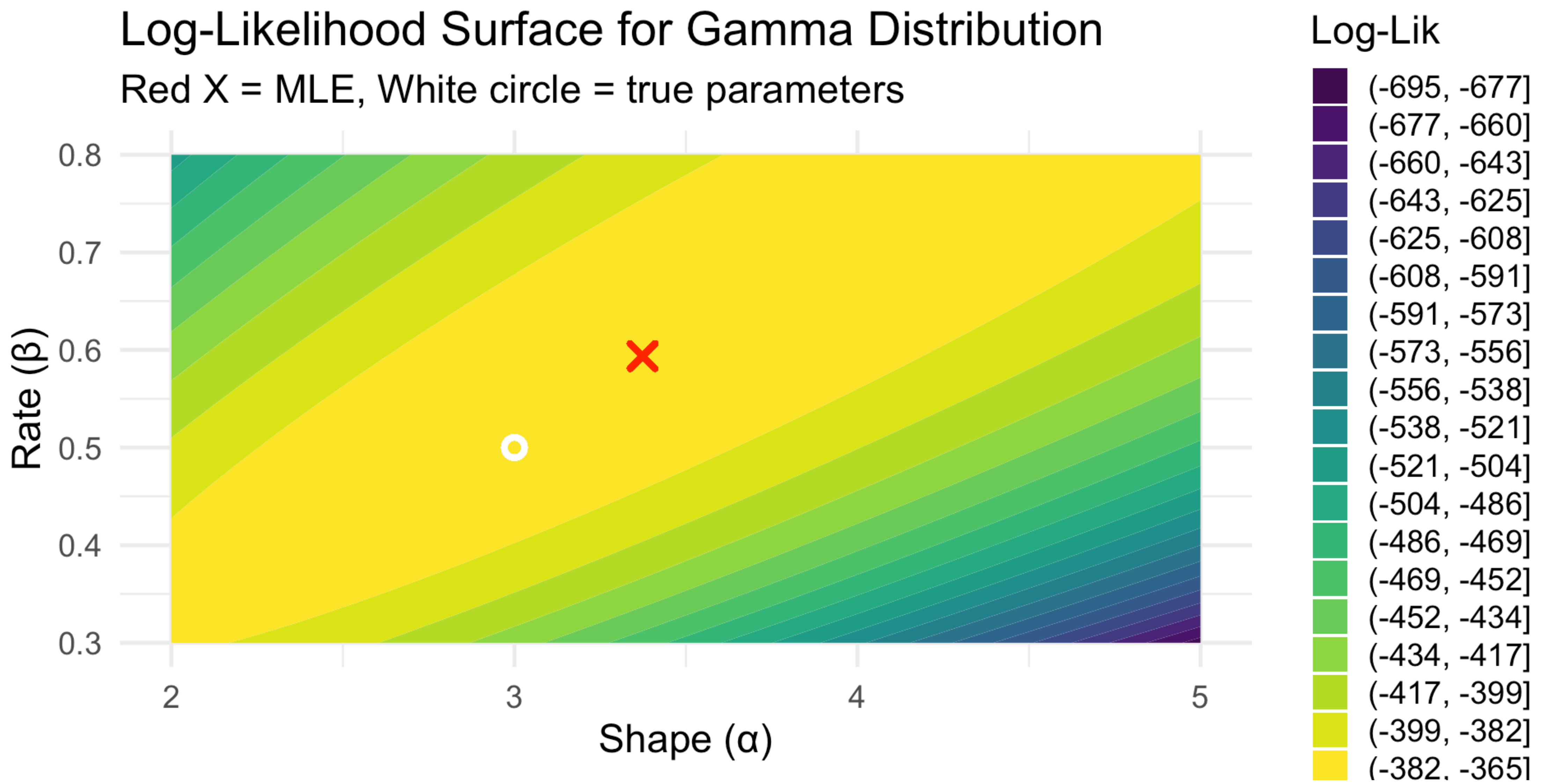
```r
# Use optim() for multi-parameter optimization
# Start with method of moments estimates as initial values
start_alpha <- mean(gamma_data)^2 / var(gamma_data)
start_beta <- mean(gamma_data) / var(gamma_data)

result <- optim(
  par = c(start_alpha, start_beta),
  fn = neg_log_lik_gamma,
  data = gamma_data,
  method = "L-BFGS-B",            # Allows bounds
  lower = c(0.001, 0.001)         # Parameters must be positive
)

cat("MLE of α:", round(result$par[1], 3), "(true: 3)\n")
```

```
MLE of α: 3.374 (true: 3)
```

```r
cat("MLE of β:", round(result$par[2], 3), "(true: 0.5)")
```

```
MLE of β: 0.594 (true: 0.5)
```

Visualizing the Gamma Log-Likelihood Surface



Using MASS::fitdistr() for Common Distributions

```r
library(MASS)
select <- dplyr::select

# fitdistr uses numerical optimization internally
gamma_fit <- fitdistr(gamma_data, "gamma")

# Compare our manual optimization with fitdistr
cat("Our optim() result:\n")
```

```
Our optim() result:
```

```r
cat("  α =", round(result$par[1], 4), ", β =", round(result$par[2], 4), "\n\n")
```

```
  α = 3.3736 , β = 0.5936
```

```r
cat("MASS::fitdistr() result:\n")
```

```
MASS::fitdistr() result:
```

```r
print(gamma_fit)
```

```
     shape          rate
  3.37359613   0.59364002
 (0.37189671) (0.07056166)
```

Same answer — `fitdistr()` is just a convenient wrapper!

Numerical Optimization: Practical Tips

Best Practices

1. **Good starting values matter!** Use MME estimates as initial guesses
2. **Check convergence:** Look at `result$convergence` (0 = success)
3. **Use appropriate bounds:** For parameters that must be positive, use `method = "L-BFGS-B"` with `lower` bounds
4. **Multiple starting points:** Try different initial values to avoid local maxima
5. **Verify your result:** Compare with `fitdistr()` or plot the log-likelihood

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```r
# Always check convergence!
cat("Convergence code:", result$convergence,
    ifelse(result$convergence == 0, "(success)", "(failed)"))
```

```
Convergence code: 0 (success)
```

Example 4: Uniform Distribution (Boundary Solution)

Setup: $X_1, \ldots, X_n \sim \text{Uniform}[0, \theta]$

Likelihood:

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} \cdot I(0 \leq x_i \leq \theta) = \frac{1}{\theta^n} \cdot I(\max(x_i) \leq \theta)$$

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Key insight: $L(\theta) = 0$ if $\theta < \max(x_i)$

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For $\theta \geq \max(x_i)$: $L(\theta) = 1/\theta^n$ is **decreasing** in θ

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Key insight: $L(\theta) = 0$ if $\theta < \max(x_i)$

For $\theta \geq \max(x_i)$: $L(\theta) = 1/\theta^n$ is **decreasing** in θ

MLE: $\hat{\theta} = \max(X_1, \ldots, X_n)$

Derivative method fails! This is a boundary maximum.

MLE vs MME for Uniform (1/2)

```r
theta <- 10
n <- 20
n_sims <- 5000

uniform_comparison <- tibble(sim = 1:n_sims) |>
  mutate(
    data = map(sim, \(s) runif(n, 0, theta)),
    mle = map_dbl(data, max),
    mme = map_dbl(data, \(d) 2 * mean(d))
  )
```

MLE vs MME for Uniform (2/2)

```r
uniform_comparison |>
  summarize(
    `MLE Bias` = mean(mle) - theta,
    `MME Bias` = mean(mme) - theta,
    `MLE Variance` = var(mle), `MME Variance` = var(mme),
    `MLE MSE` = mean((mle - theta)^2),
    `MME MSE` = mean((mme - theta)^2)
  ) |>
  pivot_longer(everything()) |>
  mutate(value = round(value, 4))
```

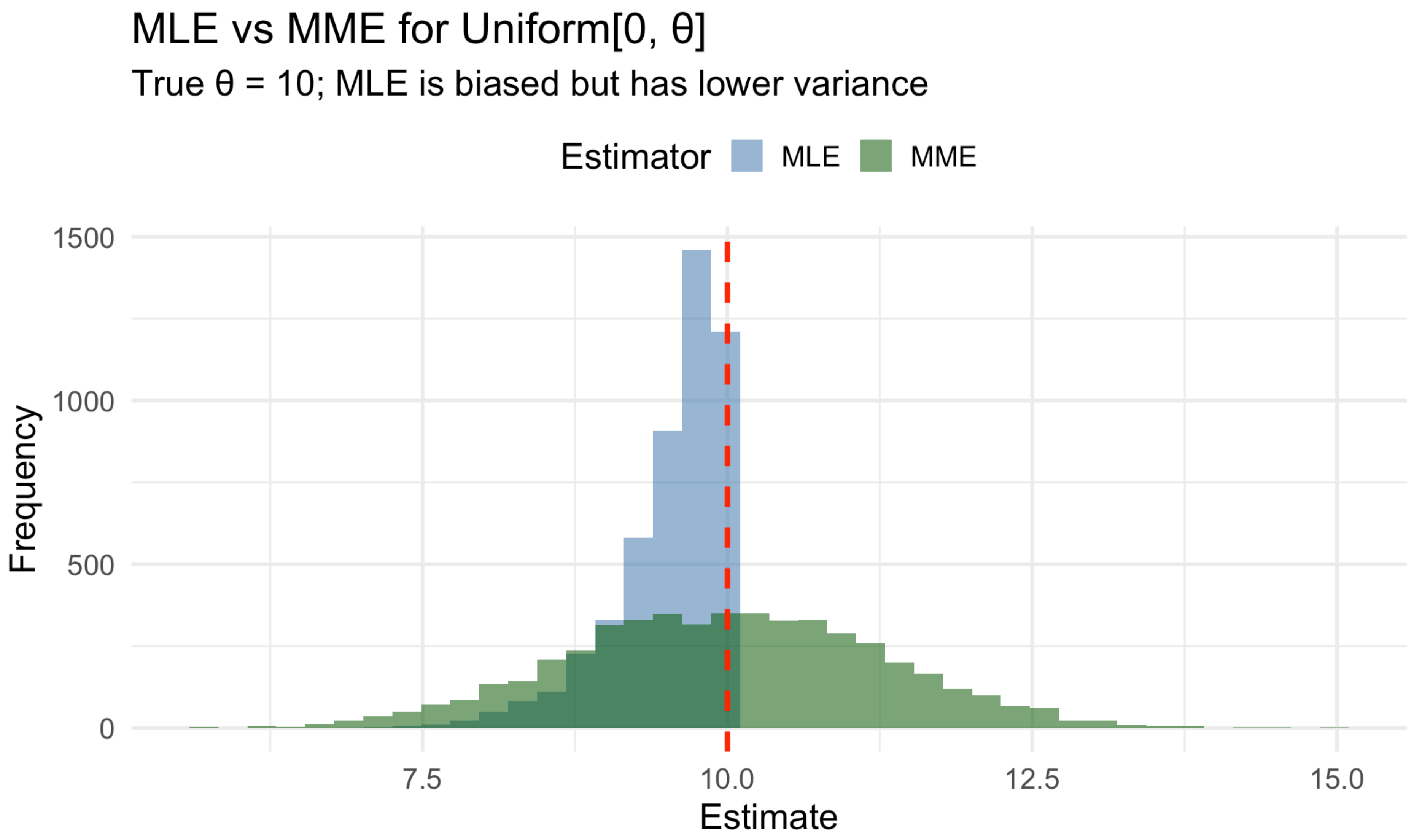
```
# A tibble: 6 × 2
  name          value
  <chr>         <dbl>
1 MLE Bias     -0.472
2 MME Bias     -0.0152
3 MLE Variance  0.198
4 MME Variance  1.65
5 MLE MSE       0.421
6 MME MSE       1.65
```

The MLE has smaller MSE despite being biased!

Visualizing MLE vs MME: Code

```r
uniform_comparison |>
  select(mle, mme) |>
  pivot_longer(everything(), names_to = "Estimator", values_to = "Estimate") |>
  mutate(Estimator = toupper(Estimator)) |>
  ggplot(aes(x = Estimate, fill = Estimator)) +
  geom_histogram(bins = 40, alpha = 0.6, position = "identity") +
  geom_vline(xintercept = theta, color = "red", linetype = "dashed", linewidth = 1.2) +
  labs(title = "MLE vs MME for Uniform[0, θ]",
       subtitle = str_glue("True θ = {theta}; MLE is biased but has lower variance"),
       x = "Estimate", y = "Frequency") +
  scale_fill_manual(values = c("steelblue", "darkgreen")) +
  theme(legend.position = "top")
```

Visualizing MLE vs MME: Result



Properties of MLEs

Property 1: Invariance

Invariance Property

If $\hat{\theta}$ is the MLE of θ, then for any function g:

$$\widehat{g(\theta)} = g(\hat{\theta})$$

The MLE of $g(\theta)$ is simply g applied to the MLE of θ.

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Example: If $\hat{p} = 0.78$ is the MLE for a proportion, then:

- MLE of odds $= p/(1-p)$: $\widehat{\text{odds}} = 0.78/0.22 = 3.55$
- MLE of $\log(p)$: $\widehat{\log p} = \log(0.78) = -0.248$

Property 2: Consistency — The Big Picture

What Does Consistency Mean?

An estimator $\hat{\theta}$ is **consistent** for θ if, as we collect more and more data, our estimate gets closer and closer to the true value.

In plain English: With enough data, we can estimate θ as accurately as we want.

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In plain English: With enough data, we can estimate θ as accurately as we want.

Think of it like this:

- With $n = 10$ observations, your estimate might be off by a lot
- With $n = 100$ observations, your estimate is probably closer
- With $n = 1000$ observations, your estimate is very close
- As $n \to \infty$, your estimate essentially equals the truth

Consistency: A Medical Analogy

Imagine estimating the true cure rate p of a new drug:

Sample Size	Your Estimate	How Close?
$n = 10$	$\hat{p} = 0.60$	Could be quite far off
$n = 100$	$\hat{p} = 0.72$	Getting closer
$n = 1,000$	$\hat{p} = 0.748$	Very close
$n = 10,000$	$\hat{p} = 0.7502$	Almost exactly right

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If the true cure rate is $p = 0.75$, a **consistent** estimator will eventually "find" this value.

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If the true cure rate is $p = 0.75$, a **consistent** estimator will eventually "find" this value.

Key Insight

Consistency is about what happens as sample size grows—it's a **long-run guarantee** about the behavior of an estimator.

Consistency: Formal Statement

Theorem: MLEs are Consistent

Under general regularity conditions, the MLE $\hat{\theta}$ satisfies:

$$\hat{\theta} \xrightarrow{P} \theta \quad \text{as } n \to \infty$$

Translation: For any tolerance $\epsilon > 0$, no matter how small:

$$P(|\hat{\theta} - \theta| < \epsilon) \to 1 \quad \text{as } n \to \infty$$

The probability that our estimate is within ϵ of the truth approaches 100%.

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The probability that our estimate is within ϵ of the truth approaches 100%.

What this does NOT mean:

- It does NOT mean $\hat{\theta}$ is unbiased
- It does NOT mean $\hat{\theta}$ is exactly right for any finite n
- It's a statement about the **limiting behavior** as n grows

Consistency via MSE → 0

Alternative Characterization

An estimator $\hat{\theta}$ is consistent if its **Mean Squared Error** goes to zero:

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] \to 0 \quad \text{as } n \to \infty$$

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Why does this work? Recall: $\text{MSE} = \text{Variance} + \text{Bias}^2$

For consistency via MSE → 0, we need both:

- $\text{Var}(\hat{\theta}) \to 0$ (estimates become more precise)
- $\text{Bias}(\hat{\theta}) \to 0$ (estimates become centered on truth)

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- $\text{Bias}(\hat{\theta}) \to 0$ (estimates become centered on truth)

Tip

This is often the easiest way to **prove** consistency — just show that MSE → 0!

Visualizing Consistency: Setup

Let's see consistency in action with the Exponential MLE.

True parameter: $\lambda = 0.5$

MLE: $\hat{\lambda} = 1/\bar{X}$

We'll simulate 1000 MLEs at each sample size and watch the distribution "shrink" around the truth.

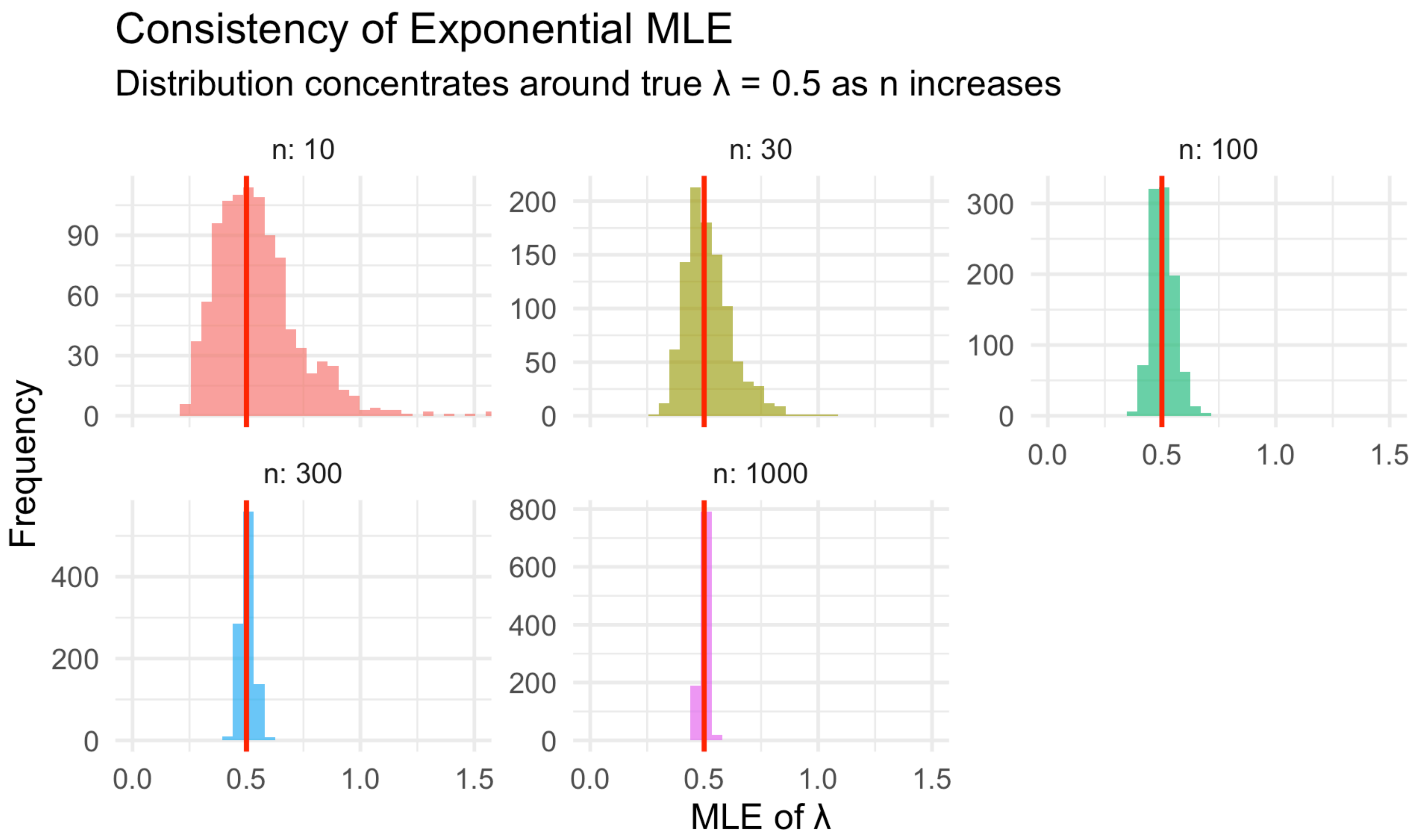
```r
true_lambda <- 0.5
sample_sizes <- c(10, 30, 100, 300, 1000)
set.seed(100)

consistency_sim <- map(sample_sizes, function(sample_n) {
  tibble(
    n = sample_n,
    sim = 1:1000,
    mle = map_dbl(sim, \(s) 1 / mean(rexp(sample_n, rate = true_lambda)))
  )
}) %>%
  list_rbind()
```

Visualizing Consistency: Code

```r
consistency_sim |>
  ggplot(aes(x = mle, fill = factor(n))) +
  geom_histogram(bins = 40, alpha = 0.7) +
  geom_vline(xintercept = true_lambda, color = "red", linewidth = 1.2) +
  facet_wrap(~n, scales = "free_y", labeller = label_both) +
  coord_cartesian(xlim = c(0, 1.5)) +
  labs(title = "Consistency of Exponential MLE",
       subtitle = "Distribution concentrates around true λ = 0.5 as n increases",
       x = "MLE of λ", y = "Frequency") +
  theme(legend.position = "none")
```

Visualizing Consistency: Result



Consistency: MSE Decomposition (1/2)

```r
# Show bias, variance, and MSE at each sample size
consistency_sim |>
  group_by(n) |>
  summarize(
    Bias = mean(mle) - true_lambda,
    Variance = var(mle),
    `Bias²` = (mean(mle) - true_lambda)^2,
    MSE = mean((mle - true_lambda)^2)
  ) |>
  mutate(across(where(is.numeric), \(x) round(x, 5)))
```

```
# A tibble: 5 × 5
      n    Bias Variance `Bias²`     MSE
  <dbl>   <dbl>    <dbl>   <dbl>   <dbl>
1    10 0.0511   0.038   0.00261 0.0406
2    30 0.0198   0.0108  0.00039 0.0112
3   100 0.00544  0.0026  0.00003 0.00263
4   300 0.00228  0.00083 0.00001 0.00083
5  1000 0.0002   0.00026 0             0.00026
```

Consistency: MSE Decomposition (2/2)

Key Observations:

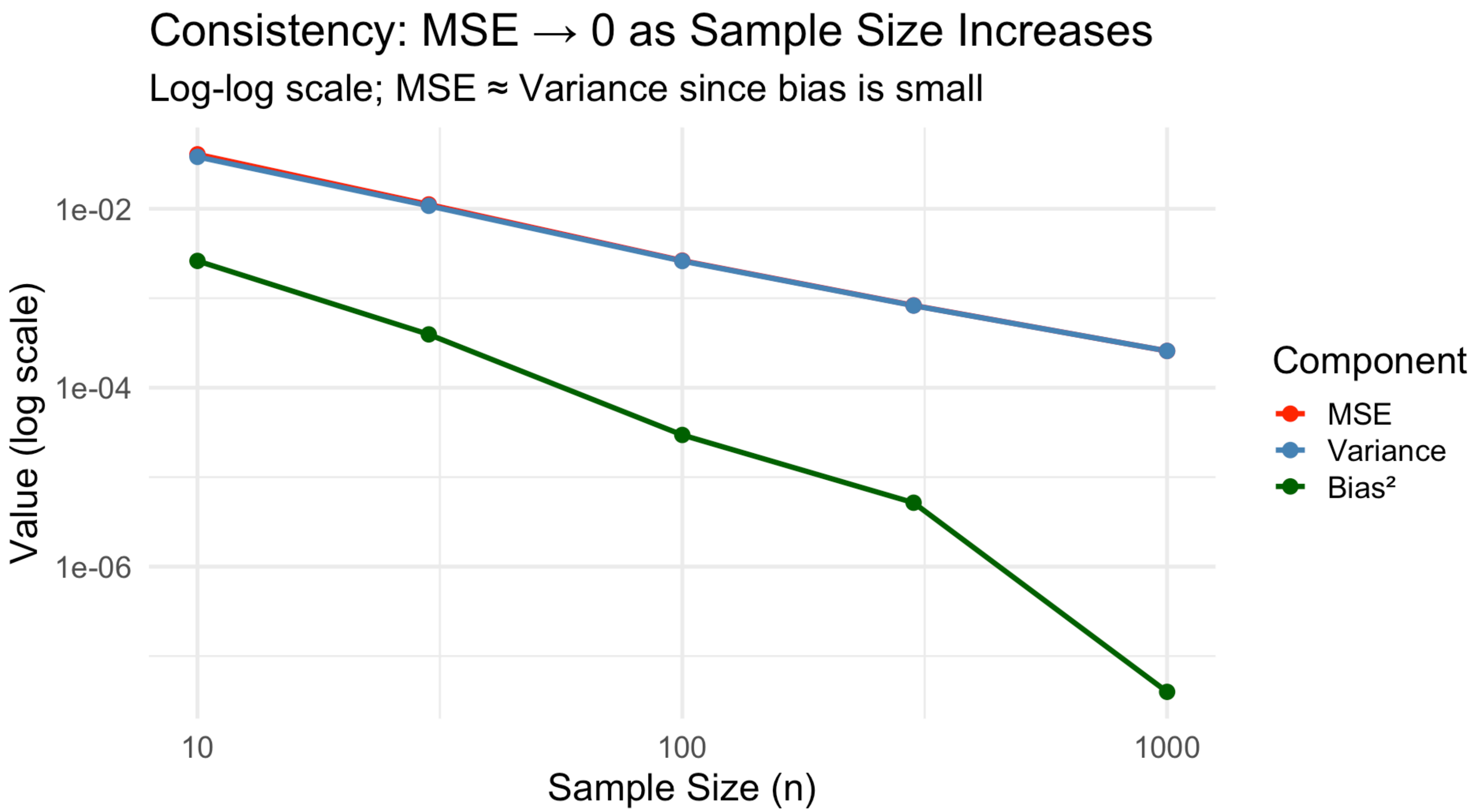
- **Bias** is small and shrinks slightly (MLE is approximately unbiased for large n)
- **Variance** shrinks dramatically as n increases
- **MSE → 0** as n increases — this proves consistency!

Visualizing MSE → 0: Code

```r
# Calculate MSE at each sample size
mse_summary <- consistency_sim |>
  group_by(n) |>
  summarize(
    MSE = mean((mle - true_lambda)^2),
    Variance = var(mle),
    Bias_sq = (mean(mle) - true_lambda)^2
  )

mse_summary |>
  pivot_longer(-n, names_to = "Component", values_to = "Value") |>
  mutate(Component = factor(Component, levels = c("MSE", "Variance", "Bias_sq"))) |>
  ggplot(aes(x = n, y = Value, color = Component)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  scale_x_log10() +
  scale_y_log10() +
  labs(title = "Consistency: MSE → 0 as Sample Size Increases",
       subtitle = "Log-log scale shows MSE shrinking toward zero",
       x = "Sample Size (n)", y = "Value (log scale)") +
  theme(legend.position = "right")
```

Visualizing MSE → 0: Result



MSE is dominated by variance (bias is negligible), and both → 0 as $n \to \infty$

Property 3: Asymptotic Normality & Efficiency

Theorem (Devore 7.4)

Under regularity conditions, for large n:

1. $\hat{\theta}$ is **approximately unbiased**: $E(\hat{\theta}) \approx \theta$
2. $\hat{\theta}$ is **approximately normal**: $\hat{\theta} \dot{\sim} N\left(\theta, \frac{1}{nI(\theta)}\right)$
3. $\hat{\theta}$ is **asymptotically efficient**: Achieves the Cramér-Rao lower bound

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Theorem (Devore 7.4)

Under regularity conditions, for large n:

1. $\hat{\theta}$ is **approximately unbiased**: $E(\hat{\theta}) \approx \theta$
2. $\hat{\theta}$ is **approximately normal**: $\hat{\theta} \dot{\sim} N\left(\theta, \frac{1}{nI(\theta)}\right)$
3. $\hat{\theta}$ is **asymptotically efficient**: Achieves the Cramér-Rao lower bound

In practice: For large samples, the MLE is approximately the MVUE!

Comparing MLE and MME

When do they agree? When do they differ?

Distribution	MLE	MME	Same?
Bernoulli(p)	$\bar{X}$	$\bar{X}$	✓
Exponential(λ)	$1/\bar{X}$	$1/\bar{X}$	✓
Normal(μ, σ^2)	$\bar{X}$, $\frac{1}{n}\sum(X_i - \bar{X})^2$	$\bar{X}$, $\frac{1}{n}\sum(X_i - \bar{X})^2$	✓
Uniform[0, θ]	$\max(X_i)$	$2\bar{X}$	✗
Weibull	Numerical	Different	✗

Comparing Estimator Properties

Property	MME	MLE
Computation	Usually simple algebra	May require optimization
Bias	Sometimes unbiased	Often biased for small n
Efficiency	Rarely efficient	Asymptotically efficient
Consistency	Usually consistent	Always consistent
Invariance	No	Yes

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Consistency	Usually consistent	Always consistent
Invariance	No	Yes

General Guidance

- Use MME for quick, rough estimates or simple distributions
- Use MLE for final analysis and complex distributions
- MLE is generally preferred when computationally feasible

Computing MLEs in R

Using built-in functions:

```r
# For distributions with known MLEs
normal_data <- rnorm(100, mean = 50, sd = 10)
cat("Normal MLE for μ:", mean(normal_data), "\n")
```

```
Normal MLE for μ: 50.91004
```

```r
cat("Normal MLE for σ:", sd(normal_data) * sqrt((100-1)/100))
```

```
Normal MLE for σ: 9.881601
```

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cat("Normal MLE for σ:", sd(normal_data) * sqrt((100-1)/100))
```

```
Normal MLE for σ: 9.881601
```

Using optimization for complex cases:

```r
# General approach: maximize log-likelihood
neg_log_lik <- function(lambda, data) {
  -sum(dexp(data, rate = lambda, log = TRUE))
}

exp_data <- rexp(50, rate = 0.5)
result <- optimize(neg_log_lik, interval = c(0.01, 5), data = exp_data)
cat("\nExponential MLE for λ:", round(result$minimum, 4))
```

```
Exponential MLE for λ: 0.4593
```

Using fitdistr for Common Distributions

```r
library(MASS)

# Fit various distributions
gamma_data <- rgamma(100, shape = 2, scale = 3)

# Method of moments (manual)
xbar <- mean(gamma_data)
s2 <- var(gamma_data) * (99/100)
alpha_mme <- xbar^2 / s2
beta_mme <- s2 / xbar

# MLE using fitdistr
gamma_mle <- fitdistr(gamma_data, "gamma")

cat("MME: α =", round(alpha_mme, 3), ", β =", round(beta_mme, 3), "\n")
```

```
MME: α = 1.979 , β = 3.144
```

```r
cat("MLE: α =", round(gamma_mle$estimate["shape"], 3),
    ", β =", round(1/gamma_mle$estimate["rate"], 3))
```

```
MLE: α = 2.015 , β = 3.087
```

Medical Application: Survival Analysis

Cancer survival times are often modeled with exponential or Weibull distributions.

```r
# Survival times (months) for 40 patients
survival_times <- c(8.2, 15.3, 12.1, 6.8, 22.5, 9.4, 18.7, 11.3, 7.5, 25.1,
                    14.2, 10.8, 5.3, 19.6, 13.4, 8.9, 16.2, 11.7, 6.1, 21.3,
                    12.8, 9.1, 17.5, 7.8, 24.2, 10.3, 14.9, 8.4, 20.1, 11.9,
                    6.5, 18.3, 13.1, 9.7, 15.8, 7.2, 22.8, 12.4, 8.1, 19.2)

# Fit exponential distribution
lambda_mle <- 1 / mean(survival_times)

# Estimated survival probability at 12 months
surv_12 <- exp(-lambda_mle * 12)

cat("MLE of λ:", round(lambda_mle, 4), "per month\n")
```

```
MLE of λ: 0.0748 per month
```

```r
cat("Estimated median survival:", round(log(2)/lambda_mle, 1), "months\n")
```

```
Estimated median survival: 9.3 months
```

```r
cat("Estimated P(survive > 12 months):", round(surv_12, 3))
```

```
Estimated P(survive > 12 months): 0.407
```

Your Turn: MLE Practice

Exercise: A diagnostic test is administered to 80 patients. 64 test positive. Assuming the number positive follows a Binomial distribution, find:

1. The MLE of the sensitivity p
2. The MLE of the odds of a positive result, $p/(1-p)$
3. The MLE of $P(\text{at least 60 positive in next 80 patients})$

Your Turn: MLE Practice

Exercise: A diagnostic test is administered to 80 patients. 64 test positive. Assuming the number positive follows a Binomial distribution, find:

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2. The MLE of the odds of a positive result, $p/(1-p)$
3. The MLE of $P(\text{at least 60 positive in next 80 patients})$

Solutions:

1. $\hat{p} = 64/80 = 0.80$
2. By invariance: $\widehat{p/(1-p)} = 0.80/0.20 = 4.0$
3. By invariance: $\widehat{P(X \geq 60)} = P(X \geq 60 | p = 0.80)$

```
1 - pbinom(59, size = 80, prob = 0.80)  # P(X >= 60)
```

```
[1] 0.8933956
```

Summary and Looking Ahead

Lesson 4 Summary

Two Methods for Constructing Estimators:

1. **Method of Moments:**
   - Equate population and sample moments
   - Simple but not always efficient
2. **Maximum Likelihood:**
   - Maximize $L(\theta)$ or $\ell(\theta)$
   - Properties: invariance, consistency, asymptotic efficiency

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Key Insight: MLEs are asymptotically MVUE—they achieve the Cramér-Rao bound for large samples!

MLE and MME Summary Table

Distribution	Parameter	MLE	MME
Bernoulli	p	$\bar{X}$	$\bar{X}$
Exponential	λ	$1/\bar{X}$	$1/\bar{X}$
Normal	μ, σ^2	$\bar{X}$, $\frac{1}{n}\sum(X_i - \bar{X})^2$	Same
Uniform[0,θ]	θ	$\max(X_i)$	$2\bar{X}$
Poisson	μ	$\bar{X}$	$\bar{X}$

Lesson 4 Practice Problems

1. **(Devore 7.25)** Derive the MLE for the parameter θ of the pdf $f(x; \theta) = (\theta + 1)x^{\theta}, 0 \leq x \leq 1$.
2. **(Devore 7.22)** Use method of moments to estimate the parameters of a Beta(α, β) distribution.
3. **(Devore 7.31)** For normal measurement errors with known mean 0, derive the MLE of the variance σ^2.

Looking Ahead

Upcoming Topics:

- Sufficient statistics (Section 7.3)
- Confidence intervals (Chapter 8)
- Hypothesis testing (Chapter 9)

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapter 7.2
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 6.

Questions?

Thank you!